

SAFETY DATA SHEET

ROVRAL® 50 WP



Version	Revision Date:	SDS Number:	Date of last issue: -
1.0	2026/02/11	50000150	Date of first issue: 2026/02/11

1. PRODUCT AND COMPANY IDENTIFICATION

Product name : ROVRAL® 50 WP

Other means of identification : None

Recommended use of the chemical and restrictions on use

Recommended use : Can be used as fungicide only.

Restrictions on use : Use as recommended by the label.
For professional users only.

Manufacturer or supplier's details

Company : FMC Taiwan Co. Ltd.

Address : 3F, No. 179, Fuhuiyuan Blvd.,
Xitun Dist., Taichung City, 40758
Taiwan

Telephone : +886 4 2254 2785

Telefax : +886 4 2254 2831

E-mail address : SDS-Info@fmc.com

Emergency telephone : For leak, fire, spill or accident emergencies, call:
00801-14-8954 (CHEMTREC)

Medical emergency:
00886 2 28757525 (in case of chemical spill, fire or poisoning
accident)

2. HAZARDS IDENTIFICATION

GHS Classification

Acute toxicity (Oral) : Category 5

Acute toxicity (Inhalation) : Category 5

Acute toxicity (Dermal) : Category 5

Carcinogenicity : Category 2

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Short-term (acute) aquatic hazard : Category 1

Long-term (chronic) aquatic hazard : Category 1

GHS label elements

Hazard pictograms :  

Signal Word : WARNING

Hazard Statements : H303 + H313 + H333 May be harmful if swallowed, in contact with skin or if inhaled.
H351 Suspected of causing cancer.
H410 Very toxic to aquatic life with long lasting effects.

Precautionary Statements : **Prevention:**
P201 Obtain special instructions before use.
P202 Do not handle until all safety precautions have been read and understood.
P273 Avoid release to the environment.
P280 Wear protective gloves/ protective clothing/ eye protection/ face protection.
Response:
P304 + P312 IF INHALED: Call a POISON CENTER/ doctor if you feel unwell.
P312 Call a POISON CENTER/ doctor if you feel unwell.
P391 Collect spillage.
Storage:
P405 Store locked up.
Disposal:
P501 Dispose of contents/ container to an approved waste disposal plant.

Other hazards which do not result in classification

None known.

3. COMPOSITION/INFORMATION ON INGREDIENTS

Substance / Mixture : Mixture

Components

Hazardous ingredients	CAS-No.	Concentration (% w/w)
iprodione (ISO)	36734-19-7	>= 30 -< 50
kaolin	1332-58-7	>= 30 -< 50

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Ethoxylated alcohols(C=12-15)	68131-39-5	>= 1 -< 2.5
Silicic acid, aluminum sodium salt	1344-00-9	>= 1 -< 10

4. FIRST AID MEASURES

General advice : Move out of dangerous area.
Show this material safety data sheet to the doctor in attendance.
Do not leave the victim unattended.

First aid measures for different exposure routes

If inhaled : If unconscious, place in recovery position and seek medical advice.
If symptoms persist, call a physician.

In case of skin contact : Wash off with soap and water.
If symptoms persist, call a physician.
Wash contaminated clothing before re-use.

In case of eye contact : Flush eyes with water as a precaution.
Remove contact lenses.
Protect unharmed eye.
Keep eye wide open while rinsing.
If eye irritation persists, consult a specialist.

If swallowed : Induce vomiting immediately and call a physician.
Keep respiratory tract clear.
Do not give milk or alcoholic beverages.
Never give anything by mouth to an unconscious person.
If symptoms persist, call a physician.
Take victim immediately to hospital.

Most important symptoms and effects, both acute and delayed : May be harmful if swallowed, in contact with skin or if inhaled.
Suspected of causing cancer.

Protection of first-aiders : First Aid responders should pay attention to self-protection and use the recommended protective clothing
Avoid inhalation, ingestion and contact with skin and eyes.
If potential for exposure exists refer to Section 8 for specific personal protective equipment.

Notes to physician : Treat symptomatically.

5. FIRE-FIGHTING MEASURES

Suitable extinguishing media : Dry chemical, CO₂, water spray or regular foam.

Unsuitable extinguishing media : Do not spread spilled material with high-pressure water streams.
High volume water jet

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| Specific hazards during fire fighting | : Do not allow run-off from fire fighting to enter drains or water courses. |
| Hazardous combustion products | : Fire may produce irritating, corrosive and/or toxic gases.
Nitrogen oxides (NOx)
Carbon oxides
Hydrogen cyanide
Hydrogen chloride
Chlorinated compounds
Sulfur oxides |
| Specific extinguishing methods | : Use a water spray to cool fully closed containers.
Remove undamaged containers from fire area if it is safe to do so.
Use extinguishing measures that are appropriate to local circumstances and the surrounding environment.
Collect contaminated fire extinguishing water separately. This must not be discharged into drains.
Fire residues and contaminated fire extinguishing water must be disposed of in accordance with local regulations. |
| Special protective equipment for fire-fighters | : Firefighters should wear protective clothing and self-contained breathing apparatus. |

6. ACCIDENTAL RELEASE MEASURES

- | | |
|---|---|
| Personal precautions, protective equipment and emergency procedures | : If it can be safely done, stop the leak.
Do not touch or walk through the spilled material.
Use personal protective equipment.
Evacuate personnel to safe areas.
Avoid dust formation.
Avoid breathing dust.
Ensure adequate ventilation.
Pick up and arrange disposal without creating dust.
Never return spills in original containers for re-use.
For disposal considerations see section 13. |
| Environmental precautions | : Prevent product from entering drains.
Prevent further leakage or spillage if safe to do so.
If the product contaminates rivers and lakes or drains inform respective authorities. |
| Methods and materials for containment and cleaning up | : Pick up and transfer to properly labeled containers without creating dust.
Move it to a safe place. |

7. HANDLING AND STORAGE

Handling

- | | |
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| Advice on protection against fire and explosion | : Avoid dust formation.
Provide appropriate exhaust ventilation at places where dust is formed. |
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Advice on safe handling : Avoid formation of respirable particles.
Do not breathe vapors/dust.
Avoid exposure - obtain special instructions before use.
Avoid contact with skin and eyes.
For personal protection see section 8.
Smoking, eating and drinking should be prohibited in the application area.
Dispose of rinse water in accordance with local and national regulations.

Storage

Conditions for safe storage : Keep container tightly closed in a dry and well-ventilated place.
Containers which are opened must be carefully resealed and kept upright to prevent leakage.
Observe label precautions.
Electrical installations / working materials must comply with the technological safety standards.

Further information on storage stability : Keep in a dry place.
No decomposition if stored and applied as directed.

8. EXPOSURE CONTROLS/PERSONAL PROTECTION

Ingredients with workplace control parameters

Components	CAS-No.	Value type (Form of exposure)	Control parameters / Permissible concentration	Basis
kaolin	1332-58-7	TWA (Respirable particulate matter)	2 mg/m ³	ACGIH
Silicic acid, aluminum sodium salt	1344-00-9	TWA (Respirable particulate matter)	1 mg/m ³ (Aluminum)	ACGIH

Biological occupational exposure limits

Contains no substances with biological exposure indices.

Personal protective equipment

Respiratory protection : Use respiratory protection unless adequate local exhaust ventilation is provided or exposure assessment demonstrates that exposures are within recommended exposure guidelines.

Filter type : Particulates type

Hand protection
Material : Wear chemical resistant gloves, such as barrier laminate, butyl rubber or nitrile rubber.

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Remarks	: The suitability for a specific workplace should be discussed with the producers of the protective gloves.
Eye protection	: Eye wash bottle with pure water Tightly fitting safety goggles
Skin and body protection	: Dust impervious protective suit Choose body protection according to the amount and concentration of the dangerous substance at the work place.
Protective measures	: Plan first aid action before beginning work with this product. Always have on hand a first-aid kit, together with proper instructions. Ensure that eye flushing systems and safety showers are located close to the working place. Wear suitable protective equipment. In the context of professional plant protection use as recommended, the end user must refer to the label and the instructions for use.
Hygiene measures	: Avoid contact with skin, eyes and clothing. Do not breathe dust. When using do not eat or drink. When using do not smoke. Wash hands before breaks and at the end of workday.

9. PHYSICAL AND CHEMICAL PROPERTIES

Physical state	: solid
Color	: gray
Odor	: slight
Odor Threshold	: No data available
pH	: 5 - 6 (1% emulsion)
Melting point/ range	: No data available
Initial boiling point and boiling range	: No data available
Flash point	: Not applicable
Evaporation rate	: Not applicable

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Flammability (solid, gas)	:	Will not burn
Self-ignition	:	No data available
Upper explosion limit / Upper flammability limit	:	No data available
Lower explosion limit / Lower flammability limit	:	No data available
Vapor pressure	:	No data available
Relative vapor density	:	No data available
Relative density	:	No data available
Density	:	1.024 g/cm ³
Bulk density	:	224 - 368 kg/m ³
Solubility(ies)		
Water solubility	:	dispersible
Solubility in other solvents	:	No data available
Partition coefficient: n-octanol/water	:	No data available
Autoignition temperature	:	No data available
Decomposition temperature	:	No data available
Viscosity		
Viscosity, dynamic	:	68 mPa.s (20 °C)
Viscosity, kinematic	:	No data available
Explosive properties	:	Not explosive
Oxidizing properties	:	The substance or mixture is not classified as oxidizing.
Surface tension	:	Not applicable
Molecular weight	:	Not applicable
Particle size	:	No data available

10. STABILITY AND REACTIVITY

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Reactivity	: No decomposition if stored and applied as directed.
Chemical stability	: No decomposition if stored and applied as directed.
Possibility of hazardous reactions	: Dust may form explosive mixture in air. No decomposition if stored and applied as directed. Dust may form explosive mixture in air.
Conditions to avoid	: Heat, flames and sparks. Avoid extreme temperatures. Avoid dust formation.
Incompatible materials	: Avoid strong acids, bases, and oxidizers.
Hazardous decomposition products	: Nitrogen oxides (NO _x) Sulfur oxides Carbon oxides Halogenated compounds

11. TOXICOLOGICAL INFORMATION

Symptoms of Overexposure : None known.

Acute toxicity

May be harmful if swallowed, in contact with skin or if inhaled.

Product:

Acute oral toxicity : LD50 (Rat, female): > 2,000 mg/kg
Method: OECD Test Guideline 425
Symptoms: Fatality, hypoactivity
Assessment: The component/mixture is minimally toxic after single ingestion.

LD50 (Rat, male and female): > 2,000 mg/kg
Method: OECD Test Guideline 401

Acute inhalation toxicity : LC50 (Rat, male and female): > 5.18 mg/l
Exposure time: 4 h
Test atmosphere: dust/mist
Method: OECD Test Guideline 403

Acute dermal toxicity : LD50 (Rat, male and female): > 2,000 mg/kg
Method: OECD Test Guideline 402
GLP: yes
Assessment: The component/mixture is minimally toxic after single contact with skin.

Components:

iprodione (ISO):

Acute oral toxicity : LD50 (Rat, male and female): > 2,000 mg/kg

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Assessment: The component/mixture is minimally toxic after single ingestion.

LD50 (Rat, male and female): 3,100 - 4,300 mg/kg
Symptoms: ataxia, Diarrhea, Nose bleeding

Acute inhalation toxicity : LC50 (Rat): > 3.29 mg/l
Exposure time: 4 h
Test atmosphere: dust/mist
Symptoms: Breathing difficulties
Assessment: The component/mixture is minimally toxic after short term inhalation.
Remarks: no mortality

Acute dermal toxicity : LD50 (Rat, male and female): > 2,000 mg/kg
Method: EPA OPP 81-2
Symptoms: Irritation
GLP: yes
Assessment: The component/mixture is minimally toxic after single contact with skin.

kaolin:

Acute oral toxicity : LD50 (Rat): > 5,000 mg/kg
Method: OECD Test Guideline 401

LD50: > 2,000 mg/kg
Method: OECD Test Guideline 420
Assessment: The substance or mixture has no acute oral toxicity

Acute inhalation toxicity : LC50 (Rat): 36 mg/l
Exposure time: 1 h
Test atmosphere: dust/mist

Acute dermal toxicity : LD50 (Rat): > 5,000 mg/kg

LD50: > 2,000 mg/kg
Method: OECD Test Guideline 402
Assessment: The substance or mixture has no acute dermal toxicity

Ethoxylated alcohols(C=12-15):

Acute oral toxicity : Acute toxicity estimate: 500 mg/kg
Method: Expert judgment

Acute inhalation toxicity : LC50 (Rat, male and female): > 1.6 mg/l
Exposure time: 4 h
Test atmosphere: dust/mist
Method: OECD Test Guideline 403
Assessment: The substance or mixture has no acute inhalation toxicity
Remarks: Based on data from similar materials

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Acute dermal toxicity : LD50 (Rat, male and female): > 2,000 mg/kg
Method: OECD Test Guideline 402
Assessment: The substance or mixture has no acute dermal toxicity
Remarks: Based on data from similar materials

Silicic acid, aluminum sodium salt:

Acute oral toxicity : LD50 (Rat, male and female): 10,000 mg/kg
Method: OECD Test Guideline 401
Remarks: Based on data from similar materials

Acute inhalation toxicity : LC0 (Rat, male and female): > 2.08 mg/l
Exposure time: 4 h
Test atmosphere: dust/mist
Method: OECD Test Guideline 403
Remarks: Based on data from similar materials
no mortality

Acute dermal toxicity : LD50 (Rabbit): > 5,000 mg/kg
Method: OECD Test Guideline 402

Skin corrosion/irritation

Based on available data, the classification criteria are not met.

Product:

Species : Rabbit
Assessment : No skin irritation
Method : OECD Test Guideline 404
GLP : yes

Components:

iprodione (ISO):

Species : Rabbit
Assessment : Not classified as irritant
Method : EPA OPP 81-5
Result : No skin irritation
GLP : yes

kaolin:

Method : OECD Test Guideline 404
Result : No skin irritation

Ethoxylated alcohols(C=12-15):

Species : Rabbit
Method : OECD Test Guideline 404
Result : No skin irritation
Remarks : Based on data from similar materials

Silicic acid, aluminum sodium salt:

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Species	:	Rabbit
Result	:	No skin irritation

Serious eye damage/eye irritation

Based on available data, the classification criteria are not met.

Product:

Species	:	Rabbit
Assessment	:	No eye irritation
Method	:	OECD Test Guideline 405
GLP	:	yes

Components:

iprodione (ISO):

Species	:	Rabbit
Result	:	slight irritation
Method	:	EPA OPP 81-4
GLP	:	yes

kaolin:

Result	:	No eye irritation
Method	:	OECD Test Guideline 405

Ethoxylated alcohols(C=12-15):

Result	:	Irreversible effects on the eye
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Silicic acid, aluminum sodium salt:

Species	:	Rabbit
Result	:	No eye irritation

Respiratory or skin sensitization

Skin sensitization

Based on available data, the classification criteria are not met.

Respiratory sensitization

Based on available data, the classification criteria are not met.

Product:

Test Type	:	Local lymph node assay (LLNA)
Species	:	mice
Method	:	OECD Test Guideline 429
Result	:	Not a skin sensitizer.

Components:

iprodione (ISO):

Test Type	:	Buehler Test
Species	:	Guinea pig

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Assessment : Not a skin sensitizer.
Method : EPA OPP 81-6
Result : Does not cause skin sensitization.
GLP : yes

kaolin:

Method : OECD Test Guideline 429
Result : Does not cause skin sensitization.

Ethoxylated alcohols(C=12-15):

Test Type : Maximization Test
Routes of exposure : Intradermal
Species : Guinea pig
Method : OECD Test Guideline 406
Result : Not a skin sensitizer.
Remarks : Based on data from similar materials

Chronic toxicity

Germ cell mutagenicity

Based on available data, the classification criteria are not met.

Product:

Genotoxicity in vitro : Test Type: Ames test
Metabolic activation: with and without metabolic activation
Method: OECD Test Guideline 471
Result: negative

Genotoxicity in vivo : Method: OECD Test Guideline 474
Result: negative

Components:

iprodione (ISO):

Genotoxicity in vitro : Test Type: Ames test
Metabolic activation: with and without metabolic activation
Result: negative

Test Type: in vitro DNA damage and/or repair study
Test system: Bacillus subtilis
Metabolic activation: with and without metabolic activation
Result: positive

Test Type: Chromosome aberration test in vitro
Test system: Chinese hamster ovary cells
Metabolic activation: with and without metabolic activation
Result: negative

Test Type: sister chromatid exchange assay
Test system: Chinese hamster ovary cells
Metabolic activation: with and without metabolic activation
Result: negative

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Genotoxicity in vivo : Test Type: Micronucleus test
Species: Mouse
Result: negative

Germ cell mutagenicity - Assessment : Weight of evidence does not support classification as a germ cell mutagen.

kaolin:

Genotoxicity in vitro : Test Type: Ames test
Method: OECD Test Guideline 471
Result: negative

Genotoxicity in vivo : Remarks: No data available

Ethoxylated alcohols(C=12-15):

Genotoxicity in vitro : Test Type: Chromosome aberration test in vitro
Method: OECD Test Guideline 473
Result: negative
Remarks: Based on data from similar materials

Test Type: Ames test
Method: OECD Test Guideline 471
Result: negative
Remarks: Based on data from similar materials

Genotoxicity in vivo : Test Type: Micronucleus test
Species: Mouse (male and female)
Application Route: Intraperitoneal injection
Method: OECD Test Guideline 474
Result: negative
Remarks: Based on data from similar materials

Test Type: Bone marrow chromosome aberration.
Species: Rat (male and female)
Method: OECD Test Guideline 475
Result: negative
Remarks: Based on data from similar materials

Silicic acid, aluminum sodium salt:

Genotoxicity in vitro : Test Type: In vitro mammalian cell gene mutation test
Method: OECD Test Guideline 476
Result: negative
Remarks: Based on data from similar materials

Genotoxicity in vivo : Test Type: chromosome aberration assay
Species: Rat (male)
Application Route: Oral
Result: negative
Remarks: Based on data from similar materials

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Carcinogenicity

Suspected of causing cancer.

Components:

iprodione (ISO):

Species	: Rat, male
Exposure time	: 2 y
	: 6.1 mg/kg bw/day
	: 12.4 mg/kg bw/day
Result	: positive
Symptoms	: Testicular effects
Target Organs	: Adrenal gland, Testes

Species	: Rat, female
Exposure time	: 2 y
	: 8.4 mg/kg bw/day
	: 16.5 mg/kg bw/day
Target Organs	: Adrenal gland

Carcinogenicity - Assessment	: Limited evidence of carcinogenicity in animal studies
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Silicic acid, aluminum sodium salt:

Species	: Rat, male and female
Application Route	: Oral
Exposure time	: 103 weeks
Result	: negative
Remarks	: Based on data from similar materials

Reproductive toxicity

Based on available data, the classification criteria are not met.

Components:

iprodione (ISO):

Effects on fetal development	: Species: Rabbit
	General Toxicity Maternal: NOAEL: 20 mg/kg bw/day
	Developmental Toxicity: NOAEL: 60 mg/kg bw/day
	Symptoms: Reduced body weight, Total Resorptions / resorption rate.

	Species: Rat
	General Toxicity Maternal: NOAEL: 20 mg/kg bw/day
	Developmental Toxicity: NOAEL: 20 mg/kg bw/day
	Symptoms: Reduced body weight, Fetal mortality.
	Target Organs: Adrenal gland

Reproductive toxicity - Assessment	: Weight of evidence does not support classification for reproductive toxicity
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Effects on fertility : Remarks: No data available

Effects on fetal development : Remarks: No data available

Ethoxylated alcohols(C=12-15):

Effects on fertility : Test Type: Two-generation study
Species: Rat, male and female
Application Route: Dermal
General Toxicity Parent: NOAEL: 250 mg/kg body weight
Fertility: NOAEC Mating/Fertility: 250 mg/kg body weight
Method: OECD Test Guideline 416
Result: negative
Remarks: Based on data from similar materials

Effects on fetal development : Test Type: reproductive and developmental toxicity study
Species: Rat
Application Route: Dermal
General Toxicity Maternal: NOEL: 100 mg/kg body weight
Embryo-fetal toxicity.: NOAEL: > 250 mg/kg body weight
Method: OECD Test Guideline 416
Result: negative
Remarks: Based on data from similar materials

STOT-single exposure

Based on available data, the classification criteria are not met.

Components:

iprodione (ISO):

Assessment : The substance or mixture is not classified as specific target organ toxicant, single exposure.

kaolin:

Remarks : No significant adverse effects were reported

STOT-repeated exposure

Based on available data, the classification criteria are not met.

Components:

iprodione (ISO):

Assessment : The substance or mixture is not classified as specific target organ toxicant, repeated exposure.

kaolin:

Assessment : The substance or mixture is not classified as specific target organ toxicant, repeated exposure.

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Repeated dose toxicity

Components:

iprodione (ISO):

Species	: Rat, male
NOAEL	: 78 mg/kg
LOAEL	: 151 mg/kg
Application Route	: Oral
Exposure time	: 90 d
Target Organs	: Reproductive organs

Species	: Rat, female
NOAEL	: 89 mg/kg
LOAEL	: 189 mg/kg
Application Route	: Oral
Exposure time	: 90 d
Target Organs	: Reproductive organs

Species	: Rat, male
NOAEL	: 28 mg/kg
LOAEL	: 207 mg/kg
Application Route	: Inhalation
Exposure time	: 28 d
Target Organs	: Adrenal gland

Species	: Rat, female
NOAEL	: 43 mg/kg
LOAEL	: 241 mg/kg
Application Route	: Inhalation
Exposure time	: 28 d
Target Organs	: Adrenal gland

kaolin:

Remarks	: No data available
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Ethoxylated alcohols(C=12-15):

Species	: Rat, male and female
NOAEL	: 500 mg/kg
Application Route	: Oral
Exposure time	: 90d
Method	: OECD Test Guideline 408
Remarks	: Based on data from similar materials

Silicic acid, aluminum sodium salt:

Species	: Rat, male and female
NOAEL	: 2,500 - 3,200 mg/kg
Application Route	: Oral
Exposure time	: 2 years
Remarks	: Based on data from similar materials

Species	: Rat, male and female
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NOAEL	:	0.0013 mg/l
Application Route	:	Inhalation
Exposure time	:	13 weeks
Remarks	:	Based on data from similar materials

Aspiration toxicity

Based on available data, the classification criteria are not met.

Components:

iprodione (ISO):

The substance does not have properties associated with aspiration hazard potential.

Further information

Product:

Remarks	:	No data available
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12. ECOLOGICAL INFORMATION

Ecotoxicity

Components:

iprodione (ISO):

Toxicity to fish	:	LC50 (Oncorhynchus mykiss (rainbow trout)): 4.1 mg/l Exposure time: 96 h
Toxicity to daphnia and other aquatic invertebrates	:	EC50 (Daphnia magna (Water flea)): 0.25 mg/l Exposure time: 48 h
Toxicity to algae/aquatic plants	:	EC50 (Scenedesmus subspicatus): > 0.5 mg/l Exposure time: 72 h
M-Factor (Acute aquatic toxicity)	:	1
Toxicity to fish (Chronic toxicity)	:	NOEC (Fish): 0.26 mg/l Exposure time: 21 d
Toxicity to daphnia and other aquatic invertebrates (Chronic toxicity)	:	NOEC (Daphnia magna (Water flea)): 0.17 mg/l Exposure time: 21 d
M-Factor (Chronic aquatic toxicity)	:	1
Toxicity to soil dwelling organisms	:	LC50 (Eisenia fetida (earthworms)): > 1,000 mg/kg Exposure time: 14 d
Toxicity to terrestrial organisms	:	LD50 (Colinus virginianus (Bobwhite quail)): > 2,000 mg/kg

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LD50 (*Apis mellifera* (bees)): > 250 µg/bee
Exposure time: 48 h
Remarks: Contact

LD50 (*Apis mellifera* (bees)): > 25 µg/bee
Exposure time: 48 h
Remarks: Oral

kaolin:

Toxicity to fish : LC50 (*Oncorhynchus mykiss* (rainbow trout)): > 100 mg/l
Exposure time: 96 h
Method: OECD Test Guideline 203

Toxicity to daphnia and other aquatic invertebrates : EC50 (*Daphnia magna* (Water flea)): > 1,000 mg/l
Exposure time: 48 h
Method: OECD Test Guideline 202

Toxicity to algae/aquatic plants : EC50 (*Raphidocelis subcapitata* (freshwater green alga)): > 100 mg/l
Exposure time: 72 h
Method: OECD Test Guideline 201

Toxicity to daphnia and other aquatic invertebrates (Chronic toxicity) : Remarks: No data available

Toxicity to microorganisms : Remarks: No data available

Ethoxylated alcohols(C=12-15):

Toxicity to fish : LC50 (*Danio rerio* (zebra fish)): > 2 mg/l
Exposure time: 96 h
Remarks: Based on data from similar materials

Toxicity to daphnia and other aquatic invertebrates : EC50 (*Daphnia magna* (Water flea)): > 2 mg/l
Exposure time: 48 h
Method: OECD Test Guideline 202
Remarks: Based on data from similar materials

Toxicity to algae/aquatic plants : EC50 (*Pseudokirchneriella subcapitata* (green algae)): > 2 mg/l
Exposure time: 72 h
Method: OECD Test Guideline 201
Remarks: Based on data from similar materials

Toxicity to fish (Chronic toxicity) : NOEC (*Pimephales promelas* (fathead minnow)): 0.11 - 0.28 mg/l
Exposure time: 30 d
Remarks: Based on data from similar materials

Toxicity to daphnia and other aquatic invertebrates (Chronic toxicity) : NOEC (*Daphnia magna* (Water flea)): 1.75 mg/l
End point: Immobilization
Exposure time: 21 d

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Remarks: Based on data from similar materials

NOEC (Daphnia magna (Water flea)): 0.77 mg/l
End point: reproduction
Exposure time: 21 d
Remarks: Based on data from similar materials

Toxicity to microorganisms : EC50 (Pseudomonas putida): > 10 g/l
Exposure time: 16.9 h
Remarks: Based on data from similar materials

Toxicity to soil dwelling organisms : LC50 (Eisenia fetida (earthworms)): > 1,000 mg/kg

Silicic acid, aluminum sodium salt:

Toxicity to fish : LL50 (Danio rerio (zebra fish)): 10,000 mg/l
Exposure time: 96 h
Method: OECD Test Guideline 203

Toxicity to daphnia and other aquatic invertebrates : EL50 (Daphnia magna (Water flea)): 10,000 mg/l
Exposure time: 48 h
Method: OECD Test Guideline 202
Remarks: Based on data from similar materials

Toxicity to algae/aquatic plants : EL50 (Desmodesmus subspicatus (green algae)): 10,000 mg/l
Exposure time: 72 h
Method: OECD Test Guideline 201

Persistence and degradability

Components:

iprodione (ISO):

Biodegradability : Result: Not readily biodegradable.

Stability in water : Degradation half life (DT50): 146 d pH: 5

Degradation half life (DT50): 0.2 d pH: 8

kaolin:

Biodegradability : Remarks: The methods for determining biodegradability are not applicable to inorganic substances.

Ethoxylated alcohols(C=12-15):

Biodegradability : Result: Readily biodegradable.
Method: OECD Test Guideline 301B
Remarks: Based on data from similar materials

Silicic acid, aluminum sodium salt:

Biodegradability : Remarks: The methods for determining biodegradability are not applicable to inorganic substances.

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Bioaccumulative potential

Components:

iprodione (ISO):

Bioaccumulation : Species: Lepomis macrochirus (Bluegill sunfish)
Bioconcentration factor (BCF): 70
Remarks: Bioaccumulation is unlikely.
See section 9 for octanol-water partition coefficient.

Partition coefficient: n-octanol/water : log Pow: 3 (20 °C)
pH: 7

kaolin:

Bioaccumulation : Remarks: Bioaccumulation is unlikely.

Partition coefficient: n-octanol/water : Remarks: Not applicable

Ethoxylated alcohols(C=12-15):

Bioaccumulation : Species: Pimephales promelas (fathead minnow)
Bioconcentration factor (BCF): 237
Exposure time: 24 d
Remarks: Based on data from similar materials

Partition coefficient: n-octanol/water : log Pow: 4.91 - 6.78 (40 °C)

Silicic acid, aluminum sodium salt:

Partition coefficient: n-octanol/water : Remarks: No data available

Mobility in soil

Components:

iprodione (ISO):

Distribution among environmental compartments : Remarks: Low mobility in soil.

kaolin:

Distribution among environmental compartments : Remarks: Low mobility in soil.

Other adverse effects

Product:

Additional ecological information : An environmental hazard cannot be excluded in the event of unprofessional handling or disposal.

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Very toxic to aquatic life with long lasting effects.

13. DISPOSAL CONSIDERATIONS

Disposal methods

- | | | |
|------------------------|---|--|
| Waste from residues | : | The product should not be allowed to enter drains, water courses or the soil.
Do not contaminate ponds, waterways or ditches with chemical or used container.
Send to a licensed waste management company. |
| Contaminated packaging | : | Empty remaining contents.
Dispose of as unused product.
Do not re-use empty containers.
Empty containers should be taken to an approved waste handling site for recycling or disposal. |

14. TRANSPORT INFORMATION

International Regulations

UNRTDG

- | | | |
|---------------------------|---|---|
| UN number | : | UN 3077 |
| Proper shipping name | : | ENVIRONMENTALLY HAZARDOUS SUBSTANCE, SOLID, N.O.S.
(Iprodione) |
| Class | : | 9 |
| Subsidiary risk | : | ENVIRONM. |
| Packing group | : | III |
| Labels | : | 9 (ENVIRONM.) |
| Environmentally hazardous | : | yes |

IATA-DGR

- | | | |
|--|---|---|
| UN/ID No. | : | UN 3077 |
| Proper shipping name | : | Environmentally hazardous substance, solid, n.o.s.
(Iprodione) |
| Class | : | 9 |
| Packing group | : | III |
| Labels | : | Miscellaneous |
| Packing instruction (cargo aircraft) | : | 956 |
| Packing instruction (passenger aircraft) | : | 956 |
| Environmentally hazardous | : | yes |

IMDG-Code

- | | | |
|----------------------|---|---|
| UN number | : | UN 3077 |
| Proper shipping name | : | ENVIRONMENTALLY HAZARDOUS SUBSTANCE, SOLID, N.O.S.
(Iprodione) |
| Class | : | 9 |
| Packing group | : | III |

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Labels	: 9
EmS Code	: F-A, S-F
Marine pollutant	: yes

Transport in bulk according to Annex II of MARPOL 73/78 and the IBC Code

Not applicable for product as supplied.

Special precautions for user

The transport classification(s) provided herein are for informational purposes only, and solely based upon the properties of the unpackaged material as it is described within this Safety Data Sheet. Transportation classifications may vary by mode of transportation, package sizes, and variations in regional or country regulations.

15. REGULATORY INFORMATION

National regulatory information

Regulations on Occupational Safety and Health Facilities	: applicable
Standards for the Storage, Cleanup, Handling and Disposal of Industrial Waste	: applicable
Regulations on Labelling and Hazard Communication of Hazardous Chemicals	: applicable
Rules on Road Traffic Safety	: applicable
Standards of Permissible Exposure Limits in Workplace	: Contains no substances with occupational exposure limit values.
Rules on the Prevention of Poisoning from Organic Solvents.	: Not applicable
Standard for the Control of Designated Hazardous and Dangerous Chemicals	: Not applicable
Establishment Standards and Safety Control Regulations for Manufacturing, Storing, Processing Public Hazardous Substances and Flammable Pressurized Gases Places	: Not applicable
Toxic and Concerned Chemical Substances Control Act	
Toxic chemical substances	: Not applicable
Concerned chemical substances	: Not applicable
Regulations for Governing Designating and Handling of Priority Management Chemicals	: applicable

The ingredients of this product are reported in the following inventories:

TCSI	: On the inventory, or in compliance with the inventory
TSCA	: Product contains substance(s) not listed on TSCA inventory.
AIIC	: Not in compliance with the inventory
DSL	: This product contains the following components that are not on the Canadian DSL nor NDSL. iprodione (ISO)
ENCS	: Not in compliance with the inventory

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ISHL	: Not in compliance with the inventory
KECI	: On the inventory, or in compliance with the inventory
PICCS	: Not in compliance with the inventory
IECSC	: On the inventory, or in compliance with the inventory
NZIoC	: Not in compliance with the inventory
TECI	: Not in compliance with the inventory

16. OTHER INFORMATION

Responsible Department	: FMC Corporation
Contact person	: Bee Huang
Revision Date	: 2026/02/11
Date format	: yyyy/mm/dd

Full text of other abbreviations

ACGIH	: USA. ACGIH Threshold Limit Values (TLV)
ACGIH / TWA	: 8-hour, time-weighted average

AIIC - Australian Inventory of Industrial Chemicals; ANTT - National Agency for Transport by Land of Brazil; ASTM - American Society for the Testing of Materials; bw - Body weight; CMR - Carcinogen, Mutagen or Reproductive Toxicant; DIN - Standard of the German Institute for Standardisation; DSL - Domestic Substances List (Canada); ECx - Concentration associated with x% response; ELx - Loading rate associated with x% response; EmS - Emergency Schedule; ENCS - Existing and New Chemical Substances (Japan); ErCx - Concentration associated with x% growth rate response; ERG - Emergency Response Guide; GHS - Globally Harmonized System; GLP - Good Laboratory Practice; IARC - International Agency for Research on Cancer; IATA - International Air Transport Association; IBC - International Code for the Construction and Equipment of Ships carrying Dangerous Chemicals in Bulk; IC50 - Half maximal inhibitory concentration; ICAO - International Civil Aviation Organization; IECSC - Inventory of Existing Chemical Substances in China; IMDG - International Maritime Dangerous Goods; IMO - International Maritime Organization; ISHL - Industrial Safety and Health Law (Japan); ISO - International Organisation for Standardization; KECI - Korea Existing Chemicals Inventory; LC50 - Lethal Concentration to 50 % of a test population; LD50 - Lethal Dose to 50% of a test population (Median Lethal Dose); MARPOL - International Convention for the Prevention of Pollution from Ships; n.o.s. - Not Otherwise Specified; Nch - Chilean Norm; NO(A)EC - No Observed (Adverse) Effect Concentration; NO(A)EL - No Observed (Adverse) Effect Level; NOELR - No Observable Effect Loading Rate; NOM - Official Mexican Norm; NTP - National Toxicology Program; NZIoC - New Zealand Inventory of Chemicals; OECD - Organization for Economic Co-operation and Development; OPPTS - Office of Chemical Safety and Pollution Prevention; PBT - Persistent, Bioaccumulative and Toxic substance; PICCS - Philippines Inventory of Chemicals and Chemical Substances; (Q)SAR - (Quantitative) Structure Activity Relationship; REACH - Regulation (EC) No 1907/2006 of the European Parliament and of the Council concerning the Registration, Evaluation,

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tion, Authorisation and Restriction of Chemicals; SADT - Self-Accelerating Decomposition Temperature; SDS - Safety Data Sheet; TCSI - Taiwan Chemical Substance Inventory; TDG - Transportation of Dangerous Goods; TECI - Thailand Existing Chemicals Inventory; TSCA - Toxic Substances Control Act (United States); UN - United Nations; UNRTDG - United Nations Recommendations on the Transport of Dangerous Goods; vPvB - Very Persistent and Very Bioaccumulative; WHMIS - Workplace Hazardous Materials Information System

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